

Guglielmo Cappellini

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Education

- PhD National PhD Program, PON R&I 2014-2020, AI for Healthcare** Rome, IT
Jan 2022 – Dec 2024
- "AI-based solution methods for PDEs with application to oncological hyperthermia". Tutored by prof. M. Vendittelli (Medical Robotics), co-funded by Medlogix Srl.
- Non-invasive temperature estimation in oncological hyperthermia (HT) treatment for decision support and optimization of dose delivery. Developed and validated a PINNs-based multiple-model adaptive observer;
 - Two-papers work presented at IEEE CDC 2023, Singapore;
 - Tutor of "Fundamentals of Automation", AY 2022/2023 and 2024/2025, with a focus on AI applications in healthcare;
 - Supervisor for 3 final projects of "Medical Robotics", and a BSc thesis student;
 - [PhD thesis](#) defended on May 30th, 2025 with Excellent classification.
- MSc Sapienza Università di Roma, Nuclear Engineering** Rome, IT
Jan 2018 – May 2021
- Thesis: "Study and characterization of radiolabeled nanovectors for application in precision molecular medicine", EQF 7, 108/110.
- Supervised by prof. R. Remetti (Radiation Protection for Nuclear Medicine).
- BSc Sapienza Università di Roma, Energy and Nuclear Engineering** Rome, IT
Sept 2015 – Nov 2017
- Thesis: "Radionuclides production and medical applications", EQF 6.
- Supervised by prof. L. Ferroni (Nuclear Energy Applications).

Experience

- Department of Computer, Automation and Management Engineering (DIAG), Sapienza Università di Roma, PostDoc** Rome, IT
Jan 2025 – present
- Development, innovation and certification of medical and non-medical devices for healthcare (Rome Technopole Spoke 6, FP4). Supervised by prof. M. Vendittelli.
- Extending the research on AI for oncological HT, considering internal measuring points and the use of graph neural networks (GNNs);
 - AI-based methods for real-time simulation of deformable tissues with application to interactive virtual reality for medical training. Preliminary results under submission;
 - Interest in GNNs and RNNs for modelling complex systems.
- Department of Radiation Oncology, Amsterdam UMC Hospital, Visiting PhD Student** Amsterdam, NL
Dec 2023 – May 2024
- Study of HT techniques and its clinical application, to investigate and to validate new solutions to improve the real time controls, and to estimate temperature at depth in superficial HT.
- Experimental emulation of HT treatment: a system to simulate perfusion, a muscle-equivalent phantom, a superficial EM applicator, and the thermometry system;
 - Transdisciplinary environment of medical physicists, engineers, and data scientists;
- Department of Nuclear Medicine, Policlinico Umberto I, Rome, MSc Thesis Intern** Rome, IT
Dec 2020 – May 2021
- "^{99m}Tc-labeled Keratin-coated gold nanoparticles for selective anticancer photothermal therapy" ([Frantellizzi, De Vincentis et. al.](#) [↗](#))
- Research on Theranostic, Radiopharmacology, Nanomedicine, and Nanophotonics;
 - Developed mathematical model for MATLAB simulation of microfluidic devices for radiolabeled nanoparticles.

Projects

AMD-STITCH: Sapienza Information-Based Technology InnovaTion Center for Health ([Reference](#))

Ontology-based data preparation on electronic medical records of Italian diabetes patients within a 13 years timeframe

- Data modeling and data cleaning, providing effective techniques for setting up a unified and shared database;
- Working within secure data environments with large-scale healthcare datasets;

ROBHOT: a robot-assisted technology for superficial hyperthermia treatments ([YouTube](#))

Grant application for EIC Accelerator (work package on AI-based temperature estimation), and for Rome Technopole.

Skills

Programming & Scientific Computing: Expert in Python for computational modelling and artificial intelligence (PyTorch, Sklearn, NumPy, SciPy, pandas), specifically with PINNs, CNNs, and GNNs; version control with GitHub; familiar with reinforcement learning, quantum computing, and high-performance computing; good understanding of Web and app development

Mathematics: Specialized in computational methods for partial differential equations (MATLAB); strong foundation in numerical analysis, calculus, linear algebra, and optimization techniques for machine learning applications

Clinical Data: Experience working with large-scale healthcare datasets in alignment with FAIR data principles

Languages: English (fluent, IELTS Academic: 7.5), Spanish, Italian (native)

Teaching & Mentorship: Passionate educator with formal pedagogical training (Percorso formativo 24 CFU in anthropological-psychological-pedagogical disciplines and teaching methodologies); committed to clear communication of complex technical concepts across disciplines

Professional Qualifications: Licensed Professional Industrial Engineer (Esame di Stato Sez. A)

Interests: Committed to deep learning applications in engineering; independent and creative mindset; interdisciplinary research background bridging nuclear engineering and medical physics

Publications

Adaptive Estimation of the Pennes' Bio-Heat Equation - I: Observer Design

Dec 2023

Cristofaro, A., Cappellini, G., Staffetti, E., Trappolini, G., Vendittelli, M.

[10.1109/CDC49753.2023.10383905](https://doi.org/10.1109/CDC49753.2023.10383905) (2023 62nd IEEE Conference on Decision and Control (CDC), Singapore)

Adaptive Estimation of the Pennes' Bio-Heat Equation - II: A NN-Based Implementation for Real-Time Applications

Dec 2023

Cappellini, G., Trappolini, G., Staffetti, E., Cristofaro, A., Vendittelli, M.

[10.1109/CDC49753.2023.10384113](https://doi.org/10.1109/CDC49753.2023.10384113) (2023 62nd IEEE Conference on Decision and Control (CDC), Singapore)

Adaptive Estimation of Pennes' Bio-Heat Equation: Observer Design and PINNs-based Implementation

Cappellini, G., Cristofaro, A., De Santis, E., Staffetti, E., Trappolini, G., Vendittelli, M.

submitted to IEEE Transactions on Control Systems Technology

Extracurricular Activities

- Independent drummer with extensive performance background across multiple genres;
- Early experiences in orchestra; co-founded bands including "[Subba and the Roots](#)" and "[La Situa](#)"; secured national touring grant (NuovoIMAIE 2018, €15k); formal jazz studies with renowned drummers (M. Guilianna, R. Gatto, M. Campanale, D. Panza, G. de Rienzo, F. Mendolia); current member of "[Dimensione Brama](#)", a multidisciplinary ensemble combining theatrical and musical performance (12th place finalists in XFactor18, 2024);
- More than 500 concerts and 3 albums released; experience in studio recording and production. [My hands on these!](#)